

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-33

Faculty

Dr. Rajasekhar N V

Associate Professor G-1

SENSE, VIT-AP University

Mail: rajasekhar.venkata@vitap.ac.in

Cabin: Faculty Area 424-C:AB-1

Transistor Model Problems

1. Transistor with fixed bias

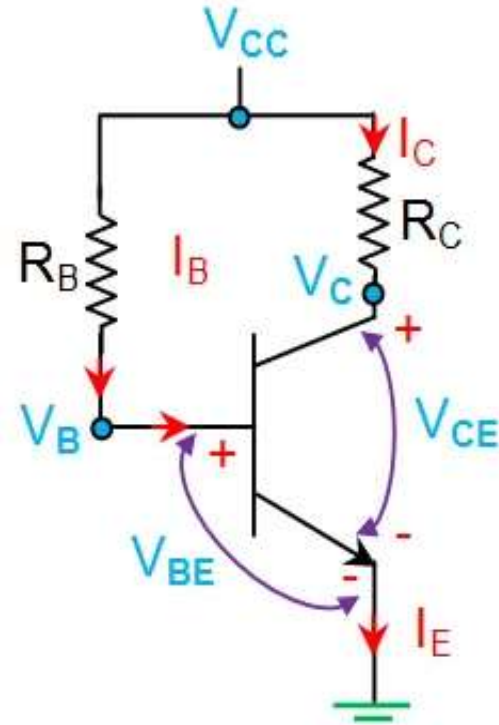
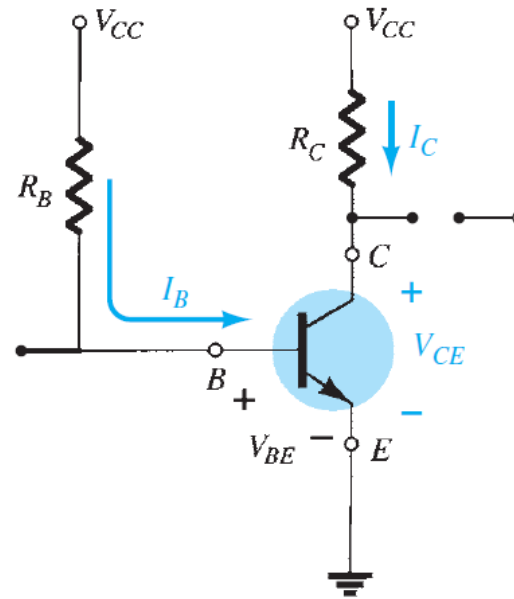
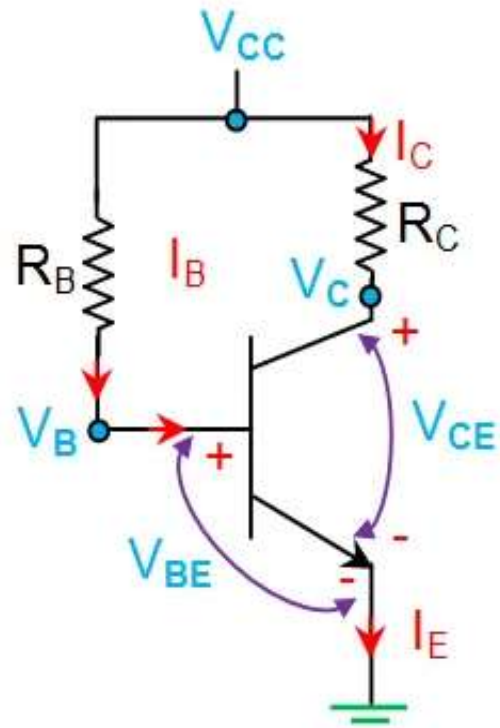
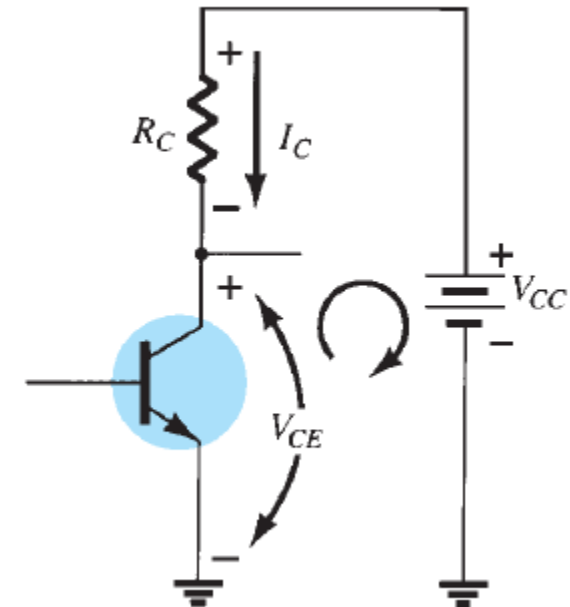
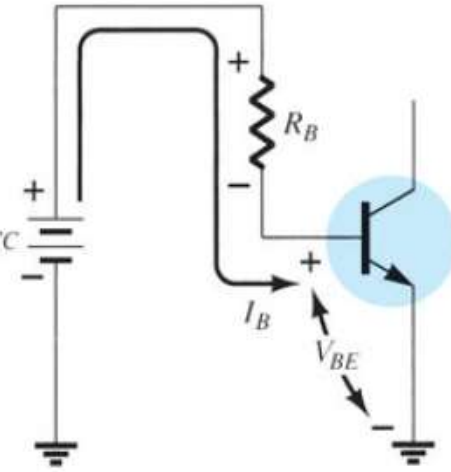


Figure: Fixed Base Bias Circuit

1. Transistor with fixed bias

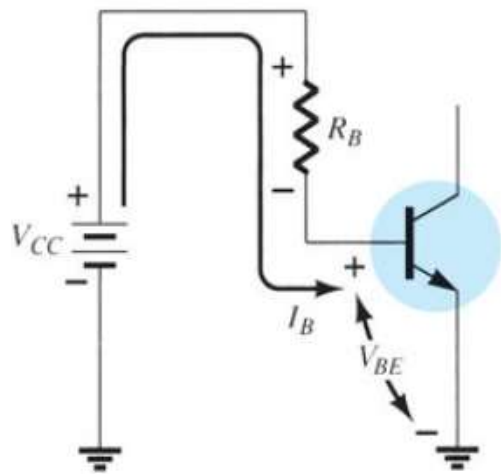


Base-emitter loop.



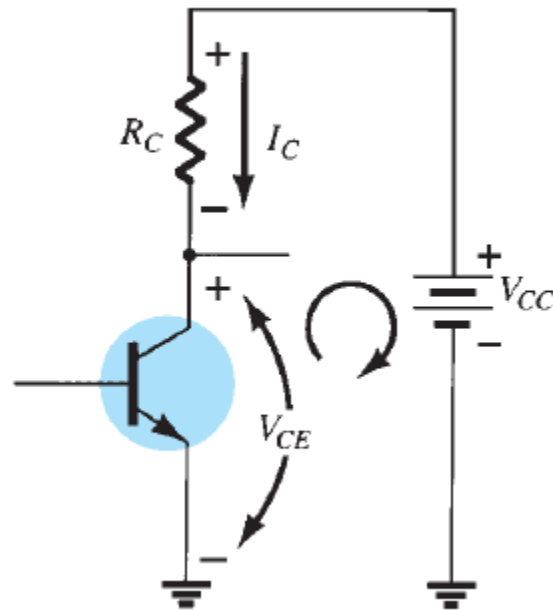
Collector-emitter loop.

Figure 1: Fixed Base Bias Circuit

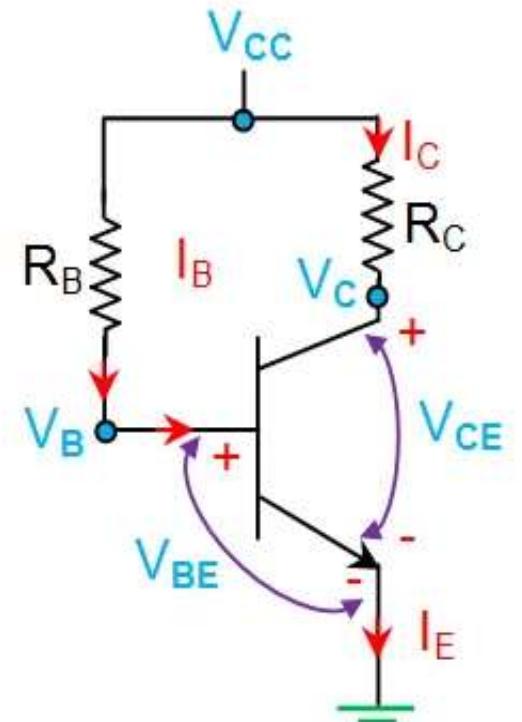


1. Base-emitter loop.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$



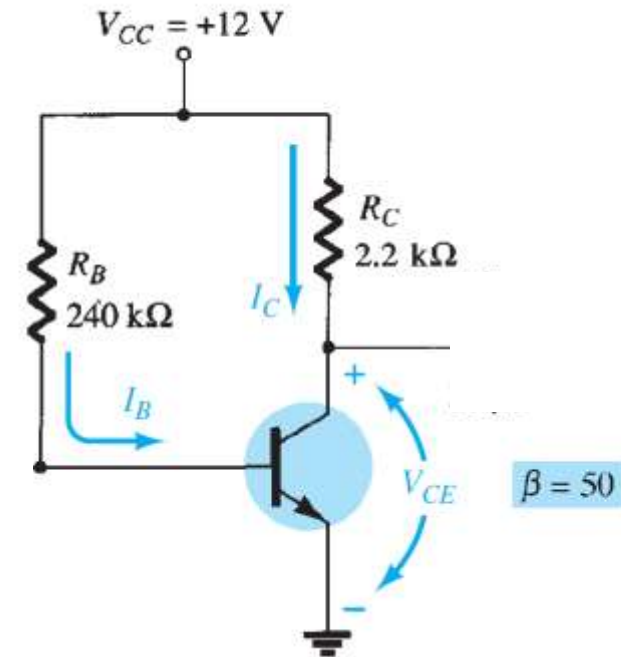
2. Collector-emitter loop.



Problem-1

Determine the following for the fixed-bias configuration of Fig .

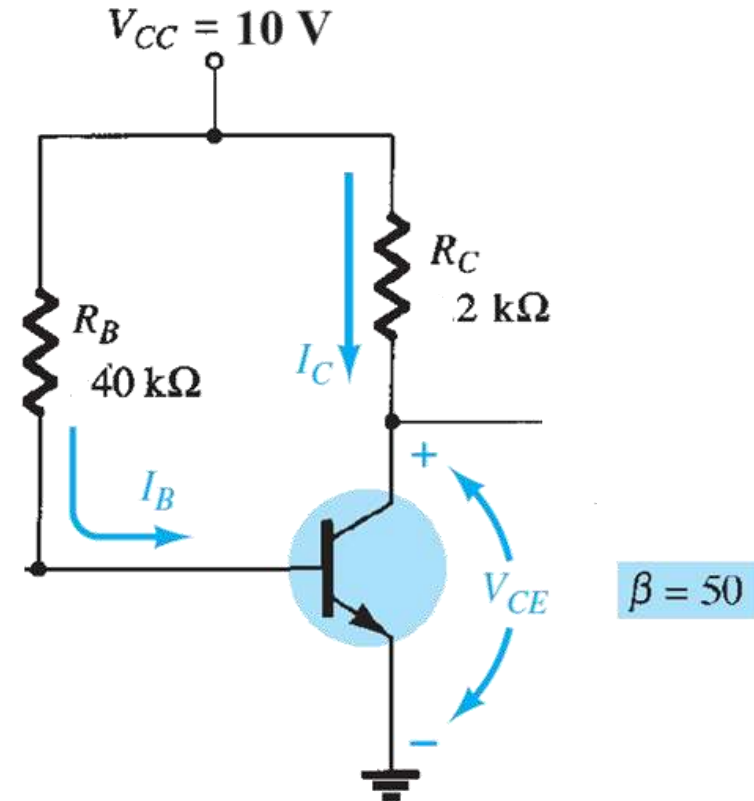
- a. I_B and I_C .
- b. V_{CE}
- c. V_B and V_C .
- d. V_{BC} .
- e. I_E



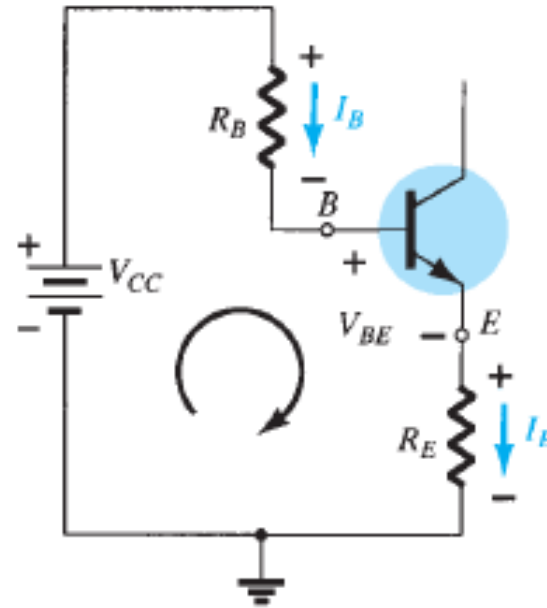
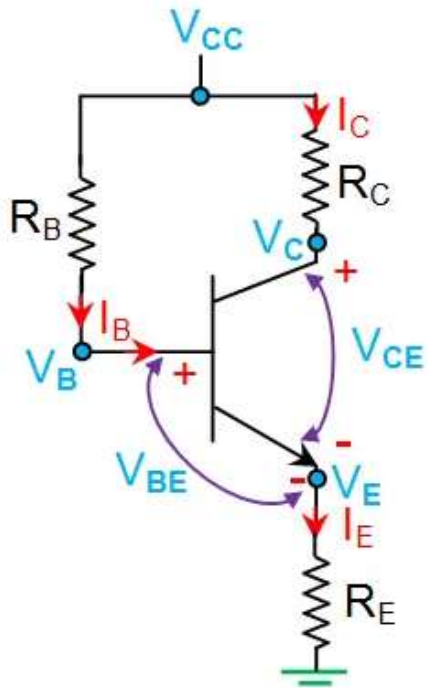
Problem-2

Determine the following for the fixed-bias configuration of Fig .

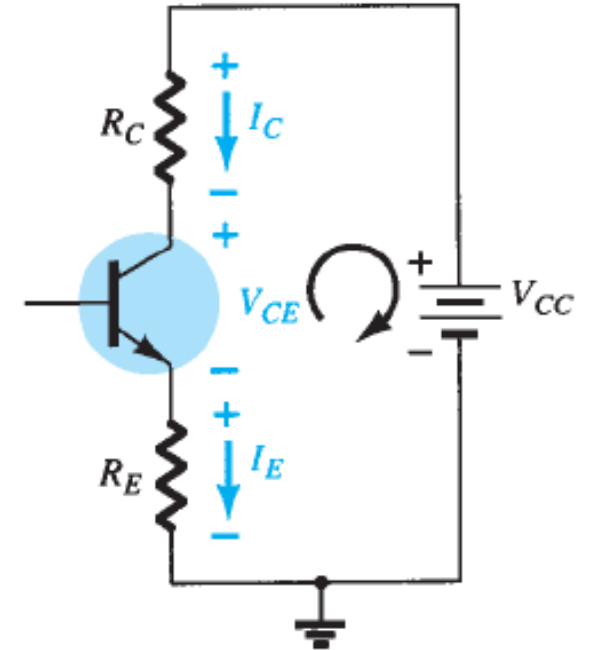
- a. I_B and I_C .
- b. V_{CE}
- c. V_B and V_C .
- d. V_{BC} .
- e. I_E



2.Fixed Bias with Emitter Resistor

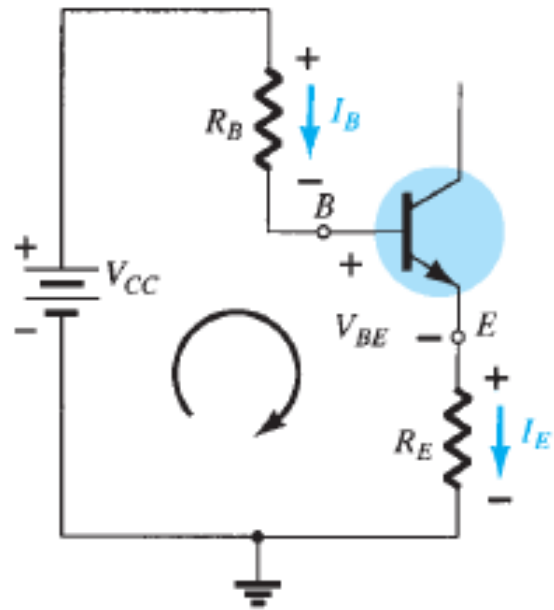


(a) Base-emitter loop.

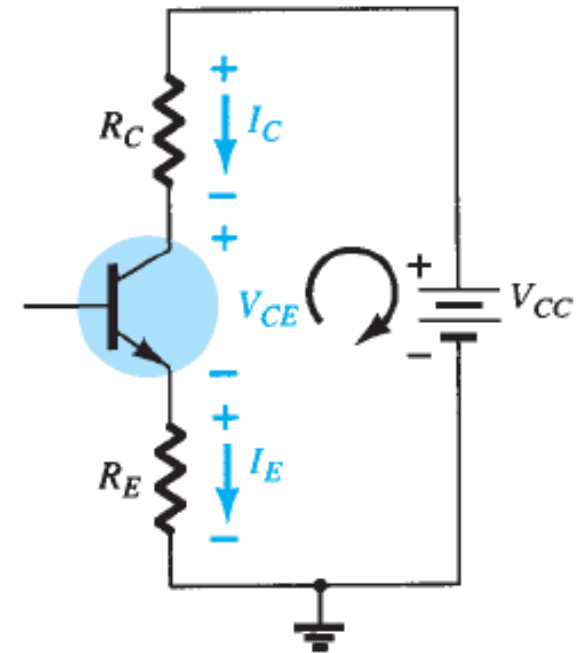


(b) Collector-emitter loop.

Figure 1: Fixed Bias with Emitter Resistor



(a) Base–emitter loop.



(b) Collector–emitter loop.

Problem

- For the emitter-bias network of Fig. below, determine:

- I_B
- I_C
- V_{CE}
- V_C
- V_E
- V_B
- V_{BC}

